

The board in **T.258/03** found that the technical character is determined by the presence or use of **technical means alone**, irrespective of purpose, even if that purpose is purely non-technical. This finding was irrespective of whether or not it can be associated with a technical effect or serves a technical purpose or function.

The board stated that what mattered, having regard to the concept of "invention" within the meaning of Art. 52(1) EPC 1973, was the presence of **technical character** which could be inferred from the physical features of an entity or the nature of an activity, or could be conferred on a non-technical activity by the use of technical means. In particular, the board held that the latter could not be considered to be a non-invention "as such" within the meaning of Art. 52(2) and (3) EPC 1973. Hence, in the board's view, activities falling within the concept of a non-invention "as such" would typically represent purely abstract concepts devoid of any technical implications. The board stated that its comparatively broad interpretation of the term "invention" in Art. 52(1) EPC 1973 would include activities which are so familiar that their technical character tends to be overlooked, such as the act of writing using pen and paper. However, this did not imply that all methods involving the use of technical means were patentable. They still have to be new, represent a non-obvious technical solution to a technical problem, and be susceptible of industrial application. The consequence was that, with regard to methods, **T.931/95** (OJ 2001, 441) was explicitly overturned by **T.258/03** (point 1 of the Headnote). In decision **T.931/95** the board had stated that **using technical means** for a purely non-technical purpose and/or for processing purely non-technical information **did not necessarily** confer technical character on any such individual steps of use or on the method as a whole. The mere occurrence of technical features in a claim did not turn the subject-matter of the claim into an invention within the meaning of Art. 52(1) EPC 1973.

In **T.38/86** (OJ 1990, 384) the board stated that the **use of technical means** for carrying out a method, partly or entirely without human intervention, which, if performed by a human being, would require him to perform mental acts, might, having regard to Art. 52(3) EPC 1973, render such a method a technical process or method and thus an invention within the meaning of Art. 52(1) EPC 1973.

1.4.4 Methods used in a technical process carried out on a physical entity

In accordance with the established case law, if a method which is not per se "technical" e.g. a mathematical method, is used in a technical process, and this process is carried out on a physical entity by some technical means implementing the method and provides as its result a change in that entity, it contributes to the technical character of the invention as a whole (**T.1814/07**).

Any claimed subject-matter defining or using technical means is an invention within the meaning of Art. 52(1) EPC (see **T.424/03** and **T.258/03**, and confirmed in **G.3/08**, OJ 2011, 10). Therefore, the mere inclusion of a computer, a computer network, a readable medium carrying a program, etc. in a claim lends technical character to the claimed subject-matter.

Features of the computer program itself (see T. 1173/97) as well as the presence of a device defined in the claim may potentially lend technical character to the claimed subject-matter (see T. 769/92, OJ 1995, 525; T. 424/03 and T. 258/03).

1.4.5 Apparatus constituting a physical entity or concrete product

An apparatus for carrying out an activity excluded as such from patentability by Art. 52(2) and (3) EPC is not itself excluded from patentability. In particular, a claim directed to a computer loaded with a program is not excluded from patentability by Art. 52(2) EPC even if the program itself would be, i.e. if the program causes no "further technical effect" when run (T. 931/95, OJ 2001, 441; see also G. 3/08, OJ 2011, 10).

In T. 931/95 (OJ 2001, 441) the board held that an apparatus constituting a physical entity or concrete product, suitable for performing or supporting an economic activity, was an invention within the meaning of Art. 52(1) EPC 1973. The board found that a computer system suitably programmed for use in a particular field, even if that was the field of business and economy, had the character of a concrete apparatus in the sense of a physical entity, man-made for a utilitarian purpose and was thus an invention within the meaning of Art. 52(1) EPC 1973. This distinction with regard to patentability between a method for doing business and an apparatus suited to perform such a method was justified in the light of the wording of Art. 52(2)(c) EPC 1973, according to which "schemes, rules and methods" were non-patentable categories in the field of economy and business, but the category of "apparatus" in the sense of "physical entity" or "product" was not mentioned in Art. 52(2) EPC 1973. This meant that, if a claim was directed to such an entity, the formal category of such a claim did in fact imply physical features of the claimed subject-matter which could qualify as technical features of the invention concerned and thus be relevant for its patentability. An apparatus constituting a physical entity or concrete product suitable for performing or supporting an economic activity, was an invention within the meaning of Art. 52(1) EPC 1973.

In T. 258/03 (OJ 2004, 575) the board was not convinced that the wording of Art. 52(2)(c) EPC 1973, according to which "schemes, rules and methods for performing mental acts, playing games or doing business" shall not be regarded as inventions within the meaning of Art. 52(1) EPC 1973, imposed a different treatment of claims directed to activities and claims directed to entities for carrying out these activities. What matters having regard to the concept of "invention" within the meaning of Art. 52(1) EPC 1973 is the presence of technical character which may be implied by the physical features of an entity or the nature of an activity, or may be conferred on a non-technical activity by the use of technical means. The board held that the apparatus was an invention within the meaning of Art. 52(1) EPC 1973, since it comprised features which were clearly technical such as a "server computer", "client computers" and a "network". The board stated that this line of reasoning was independent of the category of the claim.

In T. 2258/10 of 4 October 2011 the board found that any apparatus constituting a physical entity or concrete product has technical character. A cooking vessel clearly falls within this definition, and thus per se has technical character.

2. Non-inventions under Article 52(2) and (3) EPC

2.1. Introduction

Art. 52(2) EPC lists subject-matter or activities not regarded as inventions within the meaning of Art. 52(1) EPC. It states that in particular the following shall not be regarded as inventions within the meaning of paragraph 1; (a) discoveries, scientific theories and mathematical methods; (b) aesthetic creations; (c) schemes, rules and methods for performing mental acts, playing games or doing business, and programs for computers; (d) presentations of information.

Seeing as Art. 52(2), (3) EPC is a **limitation** to the general entitlement to patent protection laid down in Art. 52(1) EPC, it is not a matter of judicial discretion, but must have a clear legal basis in the EPC (cf. T 154/04, OJ 2008, 46; see also G 2/12, OJ 2016, A28).

According to T 930/05, the mere fact that the list of items not to be regarded as inventions in accordance with Art. 52(2) EPC 1973 was non-exhaustive, ("in particular"), indicated that there was a common criterion for exclusion which these items shared and which could serve as the basis for possible additions to the list. The enumeration of typical non-inventions in Art. 52(2) EPC 1973 covered subjects whose common feature was a lack of technical character. The catalogue of exclusions under Art. 52(2) EPC 1973, with its reference to Art. 52(1) EPC 1973, was to be regarded as a negative definition of the concept of invention.

The verification that claimed subject-matter is an invention within the meaning of Art. 52(1) EPC 1973 is in principle a **prerequisite** for the examination with respect to novelty, inventive step and industrial application since these latter requirements are defined only for inventions (cf. Art. 54(1), 56, and 57 EPC 1973) (see T 258/03, OJ 2004, 575; see also T 154/04, OJ 2008, 46).

Art. 52(3) EPC states that paragraph (2) shall exclude the patentability of the subject-matter or activities referred to therein only to the extent to which a European patent application or European patent relates to such subject-matter or activities **as such**.

The case law clearly shows that this list of "excluded" subject-matter should not be given too broad a scope of application; Art. 52(3) EPC is a bar to a broad interpretation of Art. 52(2) EPC (G 2/12 and G 2/13, OJ 2016, A28 and A29; T 154/04).

In T 154/04 (OJ 2008, 46) the board stated that the enumeration of typical non-inventions in Art. 52(2) EPC 1973 was merely a negative, non-exhaustive list of what should not be regarded as an invention within the meaning of Art. 52(1) EPC 1973. It was the clear intention of the contracting states that this list of "excluded" subject matter should not be given too broad a scope of application. Thus Art. 52(3) EPC 1973 was introduced as a bar to such a broad interpretation of Art. 52(2) EPC 1973. By referring explicitly to the "patentability of the subject-matter or activities", paragraph 3 actually enshrined the entitlement to patent protection for the non-inventions enumerated in paragraph 2 – albeit restricting this entitlement by excluding patentability "to the extent to which the European

patent application or European patent relates to such subject matter or activities as such". That no paradigm shift was intended may also be seen from the fact that e.g. Switzerland as a contracting state considered it unnecessary ("überflüssig") to include the contents of Art. 52(2) and (3) EPC 1973 in the national regulations when harmonising them with the EPC 1973 (see "Botschaft des Bundesrates an die Bundesversammlung über drei Patentübereinkommen und die Änderung des Patentgesetzes", 76.021, 24 March 1976, page 67).

Further, the board held that "technical character" was an implicit requisite of an "invention" within the meaning of Art. 52(1) EPC 1973 (requirement of "technicality"). Art. 52(2) EPC 1973 **did not exclude from patentability** any subject matter or activity having technical character, even if linked to items listed in this provision since these items are **only excluded "as such"** (Art. 52(3) EPC 1973). In examining the patentability of an invention in respect of a claim, the claim had to be construed to determine the technical features of the invention, i.e. the features which contribute to the technical character of the invention. These principles had indeed a clear and consistent basis in the EPC and in the case law of the boards of appeal and the Enlarged Board of Appeal, in particular (see also G 3/08, OJ 2011, 10; T 931/95, OJ 2001, 441; T 914/02; T 2383/10).

For further information on the technical character of the invention, see in this chapter I.A.1.1. "Technical character of an invention".

The issue of inventions with technical and non-technical subject-matter of an invention (mixed inventions) is analysed in detail in sub-chapter "Treatment of technical and non-technical features", within the chapter I.D.9. "Assessment of inventive step".

2.2. Discoveries, scientific theories and mathematical methods

Art. 52 EPC sets out in paragraph (2) (a)-(d) a non-exhaustive list of items which are not regarded as an invention. This includes "Discoveries, scientific theories and mathematical methods" in (a). These activities share the common feature that they do not aim at any direct technical result but are rather of an abstract and intellectual character (see T 22/85, OJ 1990, 12; T 854/90, OJ 1993, 669; T 338/00).

2.2.1 Discoveries and scientific theories

If a new property of a known material or article is found out, that is mere discovery and unpatentable because discovery as such has no technical effect and is therefore not an invention within the meaning of Art. 52(1) EPC. If, however, that property is put to practical use, then this constitutes an invention which may be patentable. To find a previously unrecognised substance occurring in nature is also mere discovery and therefore unpatentable. However, if a substance found in nature can be shown to produce a technical effect, it may be patentable. In addition, if a microorganism is discovered to exist in nature and to produce an antibiotic, the microorganism itself may also be patentable as one aspect of the invention. Similarly, a gene which is discovered to exist in nature may be patentable if a technical effect is revealed, e.g. its use in making a certain polypeptide or in gene therapy (Guidelines G-II, 3.1 – March 2022 version).

It was recognised in T.208/84 (OJ 1987, 14) that the fact that the idea or concept underlying the claimed subject-matter resides in a discovery does not necessarily mean that the claimed subject-matter is a discovery "as such" (G.2/88, OJ 1990, 93).

In T.1538/05 the board noted that, speaking very generally, an invention must be of a technical character and must solve a technical problem with – at least partially – technical means. It found that the claims at issue related to the discovery of a scientific theory rather than a technical teaching. The appellant claimed to have discovered a magnetic force that was hitherto unknown, and as a consequence thereof, established that theories such as the Heisenberg uncertainty principle as well as Einstein's theory of relativity were wrong. These were scientific theories or discoveries of the laws of nature different from those established. The board was in no position to determine whether these physical theories and discoveries were correct or not. Neither the claims, nor the description gave any indication of a clear technical teaching. It was clear to the board that the subject-matter claimed by the appellant was not patentable, as the appellant had not demonstrated that they were of a technical nature and that the invention could be applied to processes or devices.

2.2.2 Mathematical methods

Purely abstract or intellectual methods are not patentable. The exclusion applies if a claim is directed to a purely abstract mathematical method and the claim does not require any technical means. If a claim is directed either to a method involving the use of technical means (e.g. a computer) or to a device, its subject-matter has a technical character as a whole and is thus not excluded from patentability under Art. 52(2) and (3) EPC.

Merely specifying the technical nature of the data or parameters of the mathematical method may not be sufficient to define an invention in the sense of Art. 52(1) EPC, as the resulting method may still fall under the excluded category of methods for performing mental acts as such (Art. 52(2)(c) and (3) EPC, see Guidelines G-II, 3.3 and 3.5.1 – March 2022 version).

Even if the idea underlying an invention may be considered to reside in a mathematical method a claim directed to a technical process in which the method is used does not seek protection for the mathematical method "as such" (T.208/84, OJ 1987, 14; G.2/88, OJ 1990, 93).

The case law of the boards of appeal has established that, if a method which is not per se "technical" e.g. a mathematical method, is used in a technical process, and this process is carried out on a physical entity by some technical means implementing the method and provides as its result a change in that entity, it contributes to the technical character of the invention as a whole. Thus this feature must be taken into account when assessing inventive step (T.208/84, OJ 1987, 14; T.641/00, T.258/03, T.1814/07, OJ 2003, 352).

In T.1326/06 it was held that a method of encrypting/decrypting or signing electronic communications may be regarded as a technical method, even if it is essentially based on a mathematical method.